

Note on Dirichlet theorem

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Abstract

Over the Christmas of 2020 my PhD roommate Chang taught me number theory, Gauss's quadratic reciprocity and Dirichlet's theorem this time. The former is the jewel of elementary number theory, and the later is the beginning of analytical number theory. However, I wasn't able to digest the proof of the quadratic reciprocity, nor able to understand the Dirichlet's proof of this theorem in its full glory. In the following, we will present (1) only the statement of quadratic reciprocity and (2) Dirichlet's proof in some special cases which is enough to illustrate his general idea.

This two theorems, seems unrelated (for me) from the first sight, are actually related in the history. Legendre actually conjectured the (later so-called) Dirichlet theorem because he needed it in his proof of the law of quadratic reciprocity. However, Gauss, while not knowing how to establish the theorem about primes in arithmetic progression, found a number of different proofs of quadratic reciprocity.

1 Quadratic reciprocity

Let p, q be distinct odd primes. The congruence $x^2 \equiv q \pmod{p}$ is solvable if and only if $x^2 \equiv p \pmod{q}$ is solvable, unless $p \equiv q \equiv 3 \pmod{4}$, in which case exactly one of them is solvable.

2 Dirichlet theorem

Theorem 2.1 (Dirichlet theorem) *If q and $0 < l < q$ are relatively prime positive integers, then there are infinitely many primes of the form $kq + l$ with $k \in \mathbb{Z}$.*

In the following, we will first review some relevant results of elementary number theory, and then give the proof for three special cases $q = 3, 4, 6, l = \pm 1$. It turns out that the proofs of these three cases are the same, which comes from they share the same underlying algebraic structure. (In contrast, there are also elementary proofs of the cases $3k + 1, 4k + 1, 6k + 1$, but they are quite different and seemingly unrelated.)

2.0.1 Preliminary: a little number theory

Theorem 2.2 (Euclid's algorithm) *For any integers a and b with $b > 0$, there exist unique integers q and r with $0 \leq r < b$ such that $a = qb + r$.*

Theorem 2.3 (Bézout's identity) *If $\gcd(a, b) = d$, then there exist integers x and y such that $ax + by = d$.*

Proof Let us consider the set S of all positive integers of the form $ax + by$ where $x, y \in \mathbb{Z}$, and let $s = ax + by$ be the smallest element in S . Clearly, any divisor of a and b divides s , so we must have $d \leq s$. The goal is to show $s \leq d$ and thus $s = d$.

In order to show $s \leq d$, let us prove s is a common divisor of a, b . First, we prove s divides a , the proof for b is the same. By Euclid's algorithm, we can write $a = qs + r, 0 \leq r < s$, and thus

$$q(ax + by) = qs = a - r \Rightarrow (1 - qx)a - qyb = r \geq s \Rightarrow r = 0 \quad (1)$$

Therefore, we have $a = qs$, i.e. s divides a . Same argument applies to b . QED.

Corollary 2.4 *Two positive integers a and b are relatively prime if and only if there exist integers x and y such that $ax + by = 1$.*

Proof To show the sufficiency, note that if $ax + by = 1$ holds and d is positive and divides both a and b , then d divides 1, hence $d = 1$.

$\mathbb{Z}^*(q)$: Multiplicative group of integers modulo q Let us consider all the integers modulo q , $0, 1, \dots, q - 1$. They form an order- q abelian group under addition. Now, we consider their multiplication and ask whether they also form an abelian group under multiplication. The answer is yes but not all of them, this is because not every element has an inverse in this case. Let us examine a concrete example $q = 4$, $\mathbb{Z}(q) = \{0, 1, 2, 3\}$, among which 1 is the unit element for multiplication and 0 clearly does not have an inverse. Since $2 \times 2 = 0$, it does not have an inverse either. Thus, we have only $\{1, 3\}$ form the group.

All the elements in $\mathbb{Z}(q)$ that have an inverse under multiplication form a group called $\mathbb{Z}^*(q)$, the multiplicative group of integers modulo q

$$\mathbb{Z}^*(q) = \{n \in \mathbb{Z}(q) \mid \exists m \in \mathbb{Z}(q) \text{ s.t. } nm = 1 \pmod{q}\} \quad (2)$$

Since every abelian group is isomorphic to cyclic group or its direct product, so does the $\mathbb{Z}^*(q)$. For example, we have seen that $\mathbb{Z}^*(4) = \{1, 3\}$ is isomorphic to $\mathbb{Z}(2)$. One can also verify that $\mathbb{Z}^*(3) \simeq \mathbb{Z}(2)$, $\mathbb{Z}^*(6) \simeq \mathbb{Z}(2)$. This is the reason that the proof of Dirichlet theorem gets simplified in these three cases.

From Bézout's identity and its corollary, there is an alternative definition of $\mathbb{Z}^*(q)$. Namely, $\mathbb{Z}^*(q)$ is the group under multiplication of all integers in $\mathbb{Z}(q)$ that are relatively prime to q , which comes from

$$nm + kq = 1 \Leftrightarrow n, q \text{ are co-prime} \quad (3)$$

i.e., all the integers in $\mathbb{Z}(q)$ that are relatively prime to q form the group $\mathbb{Z}^*(q)$ under multiplication. Therefore, the order of the group $\mathbb{Z}^*(q)$ is equal to the number of integers in $\mathbb{Z}(q)$ that are co-prime to q , the later is called the Euler's function $\varphi(q)$

$$|\mathbb{Z}^*(q)| = \varphi(q) \quad (4)$$

Theorem 2.5 (Fundamental theorem of arithmetic) *Every positive integer greater than 1 can be factored uniquely into a product of primes.*

Euclid's proof of the infinitude of primes Suppose not, denote by $p_1, \dots, p_n$ the complete set of primes. Define $N = p_1 p_2 \cdots p_n + 1$. Since $N > p_i$, N cannot be prime. Therefore, N is divisible by a prime that belongs to our list. But this is also an absurdity since every prime divides the product, yet no prime divides 1. This contradiction concludes the proof.

Euclid's argument actually can be modified to deduce finer results about the infinitude of primes. For example, let us consider the following problem. All the prime numbers can be divided into two classes $3k + 1$ and $3k - 1$, at least one of them has to contain infinitely many elements. The question is, which one or both?

We can adopt Euclid's argument to prove that the class $3k - 1$ is infinite. Let assume there are only finitely many such primes and construct

$$N = 3(3k_1 - 1)(3k_2 - 1) \cdots (3k_n - 1) - 1 \quad (5)$$

N is again of the form $3p - 1$ and it must contain a prime factor $3\tilde{k} - 1$ because product of $3k + 1$ is still of the form $3k + 1$. This prime factor $3\tilde{k} - 1$ cannot be in our list because any $3k_i - 1$ divides the product $(3k_1 - 1)(3k_2 - 1) \cdots (3k_n - 1)$ but not 1. QED.

This argument can be generalized to $4k \pm 1$, $6k \pm 1$. However, it remains to determine if the class of primes of the form $3k + 1$ is infinite. The answer is yes. But the proof is quite involved, even the elementary one. More surprisingly, the elementary proof for $3k + 1$ seems to have nothing to do with the proof above, in spite of the apparent symmetry between $3k - 1$ and $3k + 1$. It does not mean that the intuition that there should exist a scheme treating $3k - 1$ and $3k + 1$ on equal footing is wrong. We just have to be clear of in which sense $3k - 1$ and $3k + 1$ are symmetric. This is first done by Dirichlet.

2.0.2 Euler's proof of the infinitude of primes

Dirichlet's idea dated back to Euler, who gave an analytical proof of the infinitude of primes. Basically, Euler managed to prove the following series diverges

$$\sum_{\text{primes}} 1/p \quad (6)$$

which cannot be true unless there are infinitely many primes. Inspired by this, Dirichlet managed to prove

$$\sum_{p=qk+\ell} 1/p \quad (7)$$

is also divergent, which implies the infinitude of the class $qk + \ell$. In this section, let us first review Euler's proof.

The key result is Euler's product formula,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} = \prod_{\text{prime}} \frac{1}{1 - p^{-s}}, \quad s > 1. \quad (8)$$

which is an analytic expression of the fundamental theorem of arithmetic. The proof, omitted here, can be done by taking some upper cutoff on both sides and pushing the cutoff to infinity. Given this formula, we can show the divergence of $\sum_{\text{primes}} 1/p$ as follows. Taking the logarithm of both sides

$$\log \zeta(s) = - \sum_p \log(1 - p^{-s}) = - \sum_p [p^{-s} + \mathcal{O}(p^{-2s})] = - \sum_p p^{-s} + \mathcal{O}(1) \quad (9)$$

The divergence follows from the fact $\zeta(1) \rightarrow \infty$.

2.0.3 Dirichlet's proof (special cases)

In this section, we present the proof for a concrete example $3k \pm 1$, which is also our question raised earlier. This proof can be easily adopted to prove $4k \pm 1$ and $6k \pm 1$ (simply replace 3 with 4 or 6). The underlying reason for such a generalization is that $\mathbb{Z}^*(3)$, $\mathbb{Z}^*(4)$ and $\mathbb{Z}^*(6)$ are all isomorphic to $\mathbb{Z}(2)$.

Dirichlet's idea to prove the infinitude of the two classes is to show the divergence of the corresponding two series

$$S_{3k+1}(s) = \sum_{p \equiv 3k+1} \frac{1}{p^s}, \quad S_{3k-1}(s) = \sum_{p \equiv 3k-1} \frac{1}{p^s} \quad (10)$$

as $s \rightarrow 1^+$. Both of them are part of the full series $\sum_p p^{-s}$, the question is how to separate them. To distinguish these two classes, Dirichlet introduces a characteristic function defined on the three classes of numbers $3k - 1, 3k, 3k + 1$

$$\chi(n) = \begin{cases} 1, & n = 3k + 1 \\ -1, & n = 3k - 1 \\ 0, & \text{otherwise} \end{cases} \quad (11)$$

One of its important property is that this function is multiplicative, $\chi(n)\chi(m) = \chi(nm)$ for all $n, m \in \mathbb{Z}$. Then we introduce Dirichlet's L -function

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \frac{\chi(p)}{p^s}} \quad (12)$$

The second equality follows from the multiplicativity of the characteristic function χ . Taking the logarithm of both sides, we find that

$$\log L(s, \chi) = \sum_p \frac{\chi(p)}{p^s} + \mathcal{O}(1) \quad (13)$$

A brute force computation shows that $L(s = 1, \chi)$ is finite, it follows that

$$\sum_p \frac{\chi(p)}{p^s} = \sum_{p \equiv 3k+1} \frac{1}{p^s} - \sum_{p \equiv 3k-1} \frac{1}{p^s} \quad (14)$$

is also finite as $s \rightarrow 1^+$. On the other hand, we know that $\sum_p 1/p^s$ is unbounded as $s \rightarrow 1^+$. Putting these two facts together, we know that

$$\sum_{p \equiv 3k \pm 1} \frac{1}{p} \quad (15)$$

must be divergent, which proves the infinitude.

The most essential ingredients of the proof above are the characteristic functions and the Dirichlet's L -function. The later completely relies on the former, its multiplicativity in particular. Though, the former seems to be a little bit exotic and un-natural. Actually, this function comes from ideas in representation theory. We will elaborate more on this point and see how it can naturally arises.

We close this section with a few comments on the generalization to other cases, which partially motivate the discussion in the next section

- Other simple cases are $q = 4$ and $q = 6$. Take $q = 4$ as an example, it also contains two classes $4k \pm 1$. Notice that $(4k - 1)^2$ belongs to the class $4k + 1$, the characteristic function can be directly adapted and all the arguments follow. The same is true for $q = 6$.
- It will be quite different if $q = 8$, for which there are four different classes $8k + 1, 8k + 3, 8k + 5, 8k + 7$. Every class squares to the class $8k + 1$ ($\mathbb{Z}^*(8)$ is isomorphic to $\mathbb{Z}(2) \times \mathbb{Z}(2)$). Simply assigning ± 1 to those classes cannot help tell them apart, and we must use a different characteristic function.
- From the above discussion, we can have a rough idea that the multiplication relation for those classes $qk + \ell$ is quite important. This is nothing but the group $\mathbb{Z}^*(q)$. Indeed, the number of classes $qk + \ell$ equals to the order of $\mathbb{Z}^*(q)$. The characteristic function is just the character of $\mathbb{Z}^*(q)$.

2.0.4 Dirichlet's proof (general cases)

In this section, let us outline Dirichlet's proof for the general cases. Details can be found in Stein's *Fourier analysis* Chapter 8.

Let us remind ourselves the goal, "If q and $0 < \ell < q$ are relatively prime positive integers, then there are infinitely many primes of the form $qk + \ell$ with $k \in \mathbb{Z}$ ", and our method, "prove that the series $\sum_{p \equiv \ell} p^{-s}$ is divergent as $s \rightarrow 1^+$ ".

The first question is how to naturally get the series $\sum_{p \equiv \ell} p^{-s}$ from the one we are already familiar with $\sum_p p^{-s}$. One naive answer is to introduce a Kronecker delta function $\delta_\ell(n)$ that picks all the numbers of the form $qk + \ell$, i.e.

$$\sum_{p \equiv \ell} \frac{1}{p^s} = \sum_p \frac{\delta_\ell(p)}{p^s} \quad (16)$$

Dirichlet's genius idea is to use the idea of Fourier analysis on the finite group $\mathbb{Z}^*(q)$ and replace the Kronecker delta with the characters of $\mathbb{Z}^*(q)$ (the great orthogonal theorem). Let us denote by $e(n)$ the character functions on $\mathbb{Z}^*(q)$, and then let χ be its extension onto $\mathbb{Z}$, called Dirichlet characters modulo q . It follows from the definition that

$$\delta_\ell(m) = \frac{1}{\varphi(q)} \sum_{\chi} \bar{\chi}(\ell) \chi(m) \quad (17)$$

Thus we have

$$\sum_{p \equiv \ell} \frac{1}{p^s} = \sum_p \frac{\delta_\ell(p)}{p^s} = \frac{1}{\varphi(q)} \sum_{\chi} \bar{\chi}(\ell) \sum_p \frac{\chi(p)}{p^s} \quad (18)$$

We can divide the above sum in two parts depending on whether or not χ is trivial

$$\frac{1}{\varphi(q)} \sum_{q \text{ not dividing } p} \frac{1}{p^s} + \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(\ell) \sum_p \frac{\chi(p)}{p^s} \quad (19)$$

the first sum on the right-hand side diverges when s tends to 1. If the rest is bounded then we are done. This requires the introduction of the L-function

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} \quad (20)$$

Similar to Euler's product formula, we have

$$\sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \frac{\chi(p)}{p^s}} \quad (21)$$

The rest of the proof will be similar to Euler's proof of the infinitude of primes. Namely, we want to take the logarithm on both sides and have

$$\log L(s, \chi) = \sum_p \frac{\chi(p)}{p^s} + \mathcal{O}(1) \quad (22)$$

For a non-trivial character, $L(1, \chi)$ is finite and non-zero, then $\log L(s, \chi)$ is bounded as $s \rightarrow 1^+$, and we can conclude that the sum on the right hand side is also finite.

However, there is one caveat in taking the logarithm. The Dirichlet character $\chi(n)$ is generally a complex number and so are the two sides of (21). Thus, they are not necessarily equal after taking the logarithm and can differ by a phase of integer multiple of 2π , i.e.

$$\log L(s, \chi) = - \sum_p \log \left(1 - \frac{\chi(p)}{p^s} \right) + 2\pi i M(s) \quad (23)$$

where $M(s)$ is an integer function. Therefore, we have to propose an appropriate definition of the logarithm such that $M(s)$ is identically zero.

For the right hand side, we define the log to be

$$\log \frac{1}{1-z} = \sum_{k=1}^{\infty} \frac{z^k}{k} \quad (24)$$

For the left hand side, we define the log to be

$$\log L(s, \chi) = \int_s^{+\infty} \frac{L'(t, \chi)}{L(t, \chi)} dt \quad (25)$$

The convergence of the integral is guaranteed by the asymptotic behavior

$$L(s, \chi) = 1 + \mathcal{O}(e^{-cs}), \quad s \rightarrow +\infty \quad (26)$$

These two definitions guarantees $M(s)$ to be continuous and equal to 0 by checking the $s \rightarrow +\infty$ limit.